"lowest set bit value"

$rax_{n+1} = rax_n + 1$

$rdx_{n+1} = \overset{\text{value of...}}{\text{lowest_set_bit}}(rax_{n+1})$

```
mov     rcx,rax     rcx := rax_0
shr     rcx,1       rcx = rcx_0 >>1
xor     rcx,rax     rcx = (rcx_0 >>1) ⊕ rcx_0
inc     rax         rax = rax_0 + 1
mov     rdx,rax     rdx := rax_0 + 1
shr     rdx,1       rdx = (rax_0 + 1) >>1
xor     rdx,rax     rdx = ((rax_0 + 1) >>1) ⊕ (rax_0 + 1)
xor     rdx,rcx
```

$rdx = \left(((rax_0+1)>>1) \oplus (rax_0+1)\right) \oplus \left((rax_0>>1) \oplus rcx_0\right)$

ie $rdx = (rax_0+1)>>1 \oplus (rax_0+1) \oplus (rax_0>>1) \oplus rax_0$

- empirically

rax → rax, rdx

0 → 1,1 0010
1 → 2,2 0011
2 → 3,1
3 → 4,4
5 → 6,2 4 → 5,1 0100
6 → 7,1 0101
7 → 8,8 1000
8 → 9,1 1001
9 → 10,2 1010
10 → 11,1 1011
11 → 12,4 1100

rax + 1

rdx = value of lowest set bit